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Subject : DSA

Unit 2, S13, SLO1 - Circular Doubly Linked Lists - Introduction

1. In a circular doubly linked list, the last node's next pointer points to the ____.

Answer: first node (head / START)

2. A circular doubly linked list contains two pointers in each node, typically named ____ and ____.

Answer: next, prev

3. In a circular doubly linked list, the first node's prev pointer points to the ____.

Answer: last node (tail)

4. The time complexity to traverse a circular doubly linked list of n nodes is ____.

Answer: $O(n)$

5. Insertion at the beginning of a circular doubly linked list requires updating ____ nodes.

Answer: two (head and tail) / 4 pointers

6. The circular nature of the list ensures that no node has a ____ next or previous pointer.

Answer: NULL

7. To delete a node from the end of a circular doubly linked list, we must update the prev pointer of the ____ and the next pointer of the ____.

Answer: head (first node), second last node

8. The traversal of a circular doubly linked list can start at ____ and loop until we reach it again.

Answer: any node (head)

9. In a circular doubly linked list, each node has exactly ____ neighbours.

Answer: two (2)

10. Compared to a singly linked list, circular doubly linked lists allow traversal in both ____.

Answer: forward and backward directions